

Cevahir Koprulu

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RESEARCH INTERESTS

Keywords: Reinforcement learning (RL), Generalization, and Curriculum Learning

My research focuses on RL: which tasks an agent trains on and what signal it learns from. My work proposes automated curriculum generation methods for heavy-tailed task distributions, temporal specifications, and cost constraints, and for the scale of batched simulators. In addition, I developed methods for reward synthesis from task-agnostic experience with a few demonstrations, and from physics-informed dynamics models with distance-aware uncertainty estimates. These works span offline RL, learning from demonstrations, unsupervised environment design, and neuro-symbolic RL, with applications in autonomous driving. Ongoing work extends curriculum generation to composing traffic behaviors and to unlearning in large language models.

EDUCATION

University of Texas at Austin, Austin, TX, USA
Electrical and Computer Engineering

M.S. 2023, PhD 2027
GPA: 3.94/4

Bilkent University, Ankara, Turkey
Electrical and Electronics Engineering

B.Sc. June 2021
GPA: 3.73/4

Honours/Awards: (i) Bilkent University Comprehensive Scholarship (ii) Scholarship of Turkish Prime Ministry (iii) Bilkent University EEE Department High Honours (iv) Ranked 24th among 2 million students in university entrance exam (v) FIRST Robotics Competition 2015: Rookie All-Star

WORK EXPERIENCE

Waymo

Software Engineering Intern (all work is under NDA) || Python, SQL, JAX

Mountain View, CA
June 2026 - August 2026

- Researched **learning from unsupervised driving data within Waymo's data flywheel** for onboard planner models, with a focus on undesirable driving behaviors. Surveyed state-of-the-art methods, implemented and compared a selection within the training infrastructure, and proposed an algorithm and data roadmap for future deployment.

Bosch Center for Artificial Intelligence

Research Intern (all work is under NDA) || Python, PyTorch

Sunnyvale, CA
May 2025 - August 2025

- Developed **CL4AD**, the *first* integration of curriculum learning into GPUDrive, a batched autonomous driving simulator, reducing wall clock time to achieve 99% success rate by 77%, more than a billion steps earlier than domain randomization, i.e., the default uniform scenario sampling method.

Honda Research Institute - US

Research Intern (all work is under NDA) || Python, PyTorch, RAY

Ann Arbor, MI
May 2024 - August 2024

- Developed an **action advising framework**, GEN2SPEC, that distills knowledge of generalist agents to train specialist agents in a continual learning setting. This framework accelerates the training of an RL agent via action advice from a transformer-based meta-RL agent that can adapt to unseen tasks.

Eatron Technologies

Engineering Intern (all work is under NDA) || Python, C++, PyTorch, ROS

Istanbul, Turkey
June 2020 - June 2021

- Designed **graph convolutional** and **self-attention**-based neural network architectures to extract spatio-temporal features of a traffic scene for trajectory prediction in a Level-2+ ADAS powered vehicle.

ROKETSAN (in collaboration with Bilkent University)

Industrial Design Project (all work is under NDA) || Python, C++, PyTorch, ROS

Ankara, Turkey
Sept 2019 - June 2020

- Developed a mobile robot that follows a human leader, combining **YOLOv3 for object detection** and **artificial potential field method for path planning** in a mapped area with unknown dynamic obstacles.

RESEARCH WORK

[Scaling Curriculum Learning For Autonomous Driving](#)

ArXiv, 2026

Cevahir Koprulu, David Paz, Feng Tao, Yuliang Guo, Xinyu Huang, Ufuk Topcu, Liu Ren

[Neural Stochastic Differential Equations for Uncertainty-Aware, Offline Reinforcement Learning](#)

ICLR, 2025

Cevahir Koprulu, Franck Djeumou, Ufuk Topcu

[Safety Prioritizing Curricula for Constrained Reinforcement Learning](#)

ICLR, 2025

Cevahir Koprulu, Thiago D. Simão, Nils Jansen, Ufuk Topcu

Dense Dynamics-Aware Reward Synthesis: Integrating Prior Experience with Demonstrations Cevahir Koprulu , Po-han Li, Tianyu Qiu, Ruihan Zhao, Tyler Westenbroek, David Fridovich-Keil, Sandeep P. Chinchali, Ufuk Topcu	L4DC, 2025
Joint Learning of Reward Machines and Policies in Environments with Partially Known Semantics Christos Verginis, Cevahir Koprulu , Sandeep Chinchali, Ufuk Topcu	Artificial Intelligence, 2024
Risk-Aware Curriculum Generation for Heavy-tailed Task Distributions Cevahir Koprulu , Thiago D. Simão, Nils Jansen, Ufuk Topcu	UAI, 2023
Reward-Machine-Guided, Self-Paced Reinforcement Learning Cevahir Koprulu , Ufuk Topcu	UAI, 2023
Act to Reason: A Dynamic Game Theoretical Driving Model for Highway Merging Applications Cevahir Koprulu , Yildiray Yildiz	CCTA, 2021

TECHNICAL SKILLS

- Programming languages: (Expert) Python, (Proficient) C++, Julia
- Frameworks: PyTorch, JAX, TensorFlow, ROS, RAY, verl, vLLM, Hugging Face Transformers

RELEVANT COURSEWORK

- **UT Austin:** Online Learning, Causality and Reinforcement Learning, Statistical Machine Learning, Learning-based Optimal Control, Game-Theoretic Modeling of Multi-Agent Systems, Program Synthesis, Cyber-Physical Systems, Reinforcement Learning, Convex Optimization, Probability and Statistics, and Complex Networks in the Real World.
- **École polytechnique fédérale de Lausanne** (Exchange - Spring 2019): Image Analysis and Pattern Recognition, Convex Optimization, Deep Learning, and Biological Modelling of Neural Networks.
- **Bilkent University:** Statistical Learning and Data Analytics, Robust Feedback Theory, Introduction to Financial Mathematics, Neural Networks, and Game Theory.

RESEARCH AND LEADERSHIP EXPERIENCES

Reinforcement Learning Reading Group <i>Organizer</i>	Austin, TX Aug 2025 - Ongoing
• RLRG at the CS Department of UT Austin is a student-run group that discusses research papers on RL.	
Center for Autonomy <i>Ph.D. Student</i>	Austin, TX Aug 2021 - Ongoing
Systems Laboratory <i>Undergraduate Researcher</i>	Ankara, Turkey Sept 2019 - July 2021
• Developed human driver models from real-traffic data that can change its reasoning level dynamically by combining level-k game theory and reinforcement learning.	
IEEE Robotics and Automation Society at Bilkent University <i>Chairman</i>	Ankara, Turkey May 2017 - June 2018
• Organized “Mühendis Kafası” in cooperation with Technology Development Foundation of Turkey: Series of sessions on Computer Vision and Deep Learning. • Gave lectures on robotics, control techniques, and related micro-controller programming: EE-101: Introduction to Robotics with Arduino.	

LANGUAGE SKILLS

- Turkish: Native proficiency
- English: TOEFL 110/120 (Fall 2020)
- French: DELF B1 (Spring 2015)

RECREATIONAL INTERESTS

I enjoy climbing, cycling, watching/playing soccer (football :)), and learning about history and psychology.